HHS Public Access
Author manuscript
J Neurooncol. Author manuscript; available in PMC 2026 March 25.

Published in final edited form as:
J Neurooncol. 2011 January ; 101(2): 179–188. doi:10.1007/s11060-010-0245-2.

Preclinical evaluation of ex vivo expanded/activated $\gamma\delta$ T cells for immunotherapy of glioblastoma multiforme

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Abstract

We have previously shown that expanded/activated $\gamma\delta$ T cells from healthy donors are cytotoxic to GBM cell lines and primary GBM explants. In this report, we examined the therapeutic effect of intracranial infusion of expanded/activated $\gamma\delta$ T cells on human minimal and established U251 tumor xenografts in athymic nude mice. Immunohistochemistry was used to determine the presence of NKG2D ligands on cell lines and tumors, and blocking studies were used to determine the effect of these ligands on $\gamma\delta$ T cell recognition. Expanded/activated $\gamma\delta$ T cells were prepared by 18-day culture in RPMI, human serum (HS), anti-CD2, IL-12, IFN-γ, and OKT-3. Anti-GBM activity of the cell product was assessed using in vitro cytotoxicity assays against the GBM cell line U251MG in suspension and in adherent culture. Ex vivo expanded/activated $\gamma\delta$ T cells were of the effector/memory phenotype, expressed Th1 cytokines, and effectively killed U251 cells in vitro. Xenografts were prepared using a U251 cell line following transfection with a firefly luciferase gene to monitor tumor progression. Mice treated with $\gamma\delta$ T cells showed slower progression of both new and established GBM xenografts versus mice that received vehicle only as determined by photon emission over time. Median survival was improved in all $\gamma\delta$ T cell treated groups between 32 and 50 days by Kaplan–Meier analysis. U251 cells expressed ULBP-2 and ULBP-3, although blocking of these reduced in vitro cytotoxicity of $\gamma\delta$ T cells to U251MG by only 33 and 25%, respectively. These studies show that expanded/activated $\gamma\delta$ T cells can mediate killing of new or established GBM xenografts, reduce tumor progression, and constitute a potentially effective novel immunotherapeutic strategy against GBM.

Keywords

Immunotherapy; Cellular therapy; Glioblastoma multiforme; Brain tumor; T cells; $\gamma\delta$ T cells

Introduction

The median survival for malignant high-grade gliomas remains a dismal 12 months and has not changed substantially over the past 50 years, despite advances in radiotherapy, chemotherapy, and surgical techniques [1]. Cellular immunotherapy remains an attractive approach, but with some isolated exceptions, classical methods that have had a measure of success in other malignancies have been disappointing when applied to GBM [2–13]. Previous approaches have failed to achieve widespread success for several reasons. Lymphokine-activated killer (LAK) cell preparations are difficult to engineer to consistently kill tumors and are short-lived in vivo [14]. In addition, NK cell, LAK cell [15], and CD4/CD8 T cell [16] killing are vulnerable to inhibitory molecules expressed by GBM cells. Cytotoxic T lymphocytes (CTL) or tumor infiltrating lymphocytes (TIL) rely principally upon adaptive immunity i.e., MHC-restricted, antigen-specific responses, presupposing that selected T cell clones recognize antigen expressed predominantly on GBM cells. However, antigens which might serve as therapeutic targets may not be ideal as they may be expressed only in a subset of tumor cells.

Although $\gamma\delta$ T cells express clonally distributed T cell receptors (TCR) and share many of the cell surface proteins and effector capabilities as the more common $\alpha\beta$ T cells, they function as innate effectors. They are not constrained by the selectivity and restriction of the MHC and therefore do not require peptide recognition and priming [17]. We have recently reported that ex vivo expanded/activated $\gamma\delta$ T cells are highly cytotoxic against several GBM cell lines and primary GBM tumor explants [18]. These findings are in agreement with several lines of evidence pointing to a broad role for $\gamma\delta$ T cells in tumor immunosurveillance from both animal [19, 20] and human [21–24] studies. Indeed, early preclinical work on $\gamma\delta$ T cell cytotoxicity against GBM has shown promise. Fujimiya [25] found that peripheral blood $\gamma\delta$ T cells activated by a combination of IL-2, IL-12 and solid phase anti-CD3 are cytotoxic against GBM targets in vitro. Yamaguchi [26] later showed that addition of IL-15 to IL-2/CD3 stimulated $\gamma\delta$ T cells resulted in an additive increase in proliferation and cytotoxicity against GBM targets. Lopez [27, 28] later showed that apoptosis-resistant $\gamma\delta$ T cells could be expanded using a CD2-initiated signaling pathway which induces a coordinated down-regulation of the IL-2Rα chain and a corresponding upregulation of the IL-15Rα chain, resulting in an increased expression of message for bcl-2 in all δ-chain phenotypes. In this report, we have adapted this method to generate expanded/activated $\gamma\delta$ T cells for adoptive transfer into the tumor environment. We show that local $\gamma\delta$ T cell infusion can slow tumor progression and improve survival in athymic nude mice bearing human U251MG GBM tumor xenografts.

Materials and methods

Mice

Athymic nude mice were obtained from Frederick Cancer Research, and maintained under specific pathogen-free conditions at the University of Alabama at Birmingham (UAB) Animal Resources Center. All procedures were reviewed and approved by the UAB Institutional Animal Care and Use Committee. Studies were performed on mice 6–8 weeks of age. The mice lack T cell function, thus allowing for durable engraftment of human tumors. Intracranial xenografts required anesthetizing the mice with an intraperitoneal injection of 132 mg/kg ketamine and 8.8 mg/kg xylazine.

Cell lines and transfectants

The human malignant glioma cell line U251MG was a gift of Darell D. Bigner, Duke Comprehensive Cancer Center, Duke University (Durham, NC). The U251ffLUC cell line was a gift from Dr. Michael Jensen, City of Hope (Duarte, CA). Authenticity of both cell lines was verified by short-tandem repeat analysis against a short-passage standard U251 cell line (UAB Heflin Genomics Center, Michael Crowley, Director). The GBM cell line was maintained in a 50:50 mixture of Dulbecco's minimum Eagle's medium (DMEM) and Ham's nutrient mixture F12 enriched with 7% heat-inactivated pooled human cGMP grade serum (Labquip; Woodbridge, ON, Canada), 2 mM L-glutamine, 100 μg/ml penicillin, 100 μg/ml streptomycin, and 0.25 μg/ml amphotericin (Mediatech; Herndon, VA, USA).

Antibodies, immunohistochemistry and flow cytometry

In all cases, the whole brain tissues were harvested from freshly killed moribund mice for the purpose of performing histological and functional assessments. Immunohistochemistry was performed on paraffin-embedded specimens of human U251 intracranial xenografts. After blocking of endogenous peroxidase, anti-MICA/B (BioLegend; San Diego, CA), IgG1 (R&D Systems; Minneapolis, MN), anti-ULBP1, anti-ULBP-2, or anti-ULBP-3 (Santa Cruz Biotechnology; Santa Cruz, CA) antibodies were applied. Binding specificity was controlled for by IgG_1 isotype controls (R&D Systems). For visualization, an anti-rabbit horseradish peroxidase was applied, followed by diaminobenzidine/H_2O_2. All sections were counterstained with hematoxylin, examined, and photographed under a Nikon microscope fitted with a Nikon D60 camera. Single cell suspensions were labeled with directly-conjugated monoclonal antibodies to MICA/B, IgG_1 (eBioscience; San Diego, CA), anti-ULBP-1, anti-ULBP-2, or anti-ULBP-3 (R&D Systems), acquired on a FACS Canto II flow cytometer and analyzed using DiVa and CellQuestPro software (BD Biosciences; San Jose, CA). Characterization and enumeration of lymphocytes pre- and post- expansion culture and the final infused cell product was performed using single-platform flow cytometry via surface labeling using directly-conjugated monoclonal antibodies (mAbs) against CD3, CD4, CD8, CD16 + 56, CD19, CD27, CD45RA, Vδ1, Vδ2, and TCR-$\gamma\delta$ and intracellular cytokines TNF-α, IFN-γ, IL-2, IL-17, GM-CSF (BD Biosciences, San Jose, CA).

Preparation of $\gamma\delta$ T cell-rich donor lymphocytes

Up to 50 ml of peripheral blood was obtained from healthy volunteers following informed consent. Mononuclear cells (MNC) were isolated using Ficoll and resuspended at a concentration of 1×10^6 cells/ml in RPMI-1640 supplemented with 15% pooled human serum. On the day of culture initiation (day 0), 1,000 U/ml human rIFN-γ (Roche); 10 U/ml human rIL-12 (Genetics Institute; Cambridge, MA obtained via NCI) and mouse anti-human CD2 mAb clone CLB 6G4 (1.0 μg/ml, mouse IgG2a, Baxter Healthcare; Deerfield, IL) were added. Twenty-four hours later (day 1), 10 ng/ml anti-CD3 mAb OKT3 (Ortho Biotech) and 300 U/ml human rIL-2 (Roche; Nutley, NJ) were added. Cultures were maintained for 18 days with the addition of fresh media as needed. IL-2 was added weekly to a concentration of 10 U/ml. The final product was depleted of CD4+ and CD8+ T cells using immunomagnetic microspheres (InVitrogen; Carlsbad, CA).

Cytotoxicity assay

U251MG target cells were labeled with ^{51}[Cr] and incubated with expanded/activated $\gamma\delta$ T cells at ratios of 0:1 (Background), 2.5:1, 5.0:1, 10:1, and 20:1 ratio of effectors/targets for 4 h at 37°C and 5% CO_2, washed ×1 and resuspended in 1 ml HBSS. Supernatants were removed to determine ^{51}[Cr] release in CPM. Data are presented at the mean percent target lysis (±SD) for triplicate determinations. Adherent cytotoxicity assays were performed as previously described [18] using the ATP-lite system (Perkin Elmer; Boston MA), with modifications to the manufacturer's procedure to accommodate cell-based killing. U251MG target cells were washed, trypsinized, counted and plated in 96 well plates (1,000 cells per well) and incubated overnight. Effectors (expanded/activated $\gamma\delta$ T cells) were then added to the wells in a constant volume. Wells containing only effector cells were included as an internal control. After 4 h incubation 50 μl of ATP lysis solution was added to each well followed by the addition of 50 μl ATP after 5 min. Plates were read on a luminometer (TopCount NXT; Packard, Meriden, CT) and data reported as counts per second (CPS). The mean CPS value of effectors was subtracted from the mean CPS value of the respective targets and divided by the mean control CPS value to give a percentage of viability for each group.

Tumor xenografts and immunotherapy

The U251MG cell line was transfected to express transfected to express the firefly luciferase Zeocin-resistance fusion protein to monitor tumor growth and response to therapy. Tumor cells suspended in 5% methyl cellulose in serum-free medium were drawn into a 250 μl Hamilton gas-tight syringe mounted in a Chaney repeating dispenser and fitted with a 30G ½-inch needle that has a plastic sleeve attached to mark the depth of injection as 2.5 mm from the middle of the bevel opening. Under an operating microscope, the fascia on the skull of the anesthetized mouse were scraped off with a sterile wooden dowel and a 0.5 mm burr hole made 2 mm to the right of the midline suture and 1 mm caudal to the coronal suture. The syringe is inserted into a Kopf stereotactic electrode clamp mounting bracket attached to an electrode manipulator (#1460) mounted on a Kopf stereotactic frame electrode A–P zeroing bar (#1450). Each mouse was positioned on the stereotactic frame and the needle inserted to the depth marker into the right cerebral hemisphere. Approximately 90–120 s after injection of 5 μl, the needle was slowly withdrawn over the next minute. The burr hole was plugged with sterile bone wax and skin is closed with Tissu-Mend surgical adhesive and the mice are allowed to recover in a fresh, sterile cage that is warmed on a heating pad.

Since there is no suitable model of post-resection intracranial immunotherapy in mice, we employed a minimal residual disease (MRD) model and an established disease model. For the MRD model, 2×10^5 U251 cells were injected as described above followed by either 1.25×10^6 expanded/activated $\gamma\delta$ T cells or saline through the same burr hole. For the established model, 2×10^5 U251 cells were injected and imaging studies conducted 3–5 days later to confirm establishment of viable tumor. At 1 and 2 weeks following tumor placement, either 1.25×10^6 expanded/activated $\gamma\delta$ T cells or saline was injected using the same stereotactic settings at weekly intervals.

Imaging procedures

Tumor growth was monitored at weekly intervals. Mice were injected i.p. with 25 mg/kg D-luciferin (Xenogen Corp., Alameda, CA, USA) in PBS. After 10 min the mice were anesthetized and placed in a light-tight box under the cryogenically cooled IVIS camera (Xenogen Corp.). Bioluminescence images are recorded between 10 and 20 min post luciferin administration. The bioluminescence intensity is quantified with the LivingImage software (Xenogen) and signal intensity is quantified as the sum of detected photons per second within the region of interest using the LivingImage software package.

Statistical analysis

Descriptive statistics were used to express data from cytotoxicity assays. For bioluminescent data analysis, the region of interest encompassing the intracranial space was drawn using Living Image software, and total photons per second in the region of interest were recorded and analyzed using ANOVA and the Gehan's Wilcoxon test differences between groups. For mouse xenograft therapeutic studies using established tumors, photon output collected for control (saline-treated) tumor-bearing mice were compared to that from $\gamma\delta$ T cell-treated tumor-bearing mice. If tumors failed to engraft prior to the first $\gamma\delta$ T cell treatment as determined by imaging analysis, those mice were excluded from response and survival studies. The major endpoint in this study was animal survival; moribund animals that became unresponsive to mild external stimuli were euthanized and the date of death estimated to be the day the animal is killed. Kaplan–Meier analysis was used to estimate survival from tumor induction. The Mantel–Haenszel log-rank test was used to test for differences observed between treatment groups.

RESULTS

Phenotypic and function of expanded/activated $\gamma\delta$ T cells

After 14 days in culture, the $\gamma\delta$ T cell population increased between 100 and 600 fold and were enriched by immunomagnetic depletion of CD4+ and CD8+ cells to a final cell product of ≥65% $\gamma\delta$ T cells with a small population of NK and $\alpha\beta$+CD4−CD8− T cells remaining (Fig. 1a, b, c). The $\gamma\delta$ T cell population was predominately Vδ2+, although the Vδ1+ population was also expanded (Fig. 2a). Vδ1+ cells were predominately from the CD45+CD27− T-effector/memory RA phenotype (TEMRA) while Vδ2+ cells were effector/memory phenotype. A small population of naïve (CD45+CD27+) cells were seen in both Vδ1+ and Vδ2+ populations (Fig. 2b). The cells produced Th1 cytokines IFN-γ and TNF-α as well as GM-CSF but did not show production of IL-2 or IL-17 (Fig. 3). Expanded/activated $\gamma\delta$ T cells readily lysed U251MG cells in suspension culture (Fig. 4a) and in co-culture with adherent U251 cells (Fig. 4b–c).

Regression of GBM xenografts and improved survival after the intracranial adoptive transfer of expanded/activated $\gamma\delta$ T cells: The in vivo anti-tumor effect of adoptively transferred expanded/activated $\gamma\delta$ T cells was examined using an orthotopic xenograft model using

U251MG cells transfected to express the firefly luciferase Zeocin-resistance fusion protein ($U251^{ffluc}$). In this model, 5×10^5 $U251^{ffluc}$ tumor cells were stereotactically implanted into the forebrain of athymic nude mice. In the MRD model detailed above, 1.25×10^6 $\gamma\delta$ T cells were injected at the same time as the tumor cells. Figure 5a shows photon emission scanning of mice 2 weeks following intracranial injections of saline (control) or injection of $\gamma\delta$ T cells. Tumor mass as measured by photon output is detailed in Fig. 5b and showed significantly slower progression of tumor in mice treated with $\gamma\delta$ T cells ($P = 0.0004$). Median survival of $U251^{ffluc}$-injected mice treated $\gamma\delta$ T cells as shown in Fig. 5c was 49 days with 2/10 75-day survivors as compared with control mice treated only with the saline vehicle and 0/10 75-day survivors ($P < .001$).

We also examined the effect of adoptively transferred expanded/activated $\gamma\delta$ T cells on established U251 tumors. In this model, 5×10^5 $U251^{ffluc}$ tumor cells were injected as described above. Tumor growth was confirmed by photon emission imaging 5 days following injection. On day +7 and day +14 following injection of tumor cells, 1.25×10^6 $\gamma\delta$ T cells (or PBS) were stereotactically injected through the same burr hole as the tumor using the identical depth setting. Photon output over time decreased in the both the group that received a single injection of $\gamma\delta$ T cells and two injections of $\gamma\delta$ T cells when compared to PBS-treated controls although the differences did not reach significance ($P = 0.057$ and $P = 0.114$ respectively). As shown in Fig. 6, median survival of $U251^{ffluc}$-injected mice treated with one or two injections $\gamma\delta$ T cells weekly following tumor placement was marginally improved at 32 and 50 days respectively as compared 26 days for control mice ($P = 0.1$ for both groups). Long-term (180 day) survival was also noted for 4/10 survivors in the group receiving a single injection of $\gamma\delta$ T cells and 3/10 receiving two injections. One PBS-treated control mouse showed slow establishment and progression of tumor growth and survived to 170 days.

Expression of NKG2D ligands on U251MG cells and tumors

It has been previously reported that U251MG cells express mRNA for several NKG2D ligands [29]. We examined surface expression of these ligands on U251MG cells growing in culture by flow cytometry and on solid tumor xenografts by immunohistochemistry. NKG2D ligands ULBP-2 and ULBP-3 were expressed on cultured cells and on solid tumor xenografts while MIC A/B and ULBP-1 expression was not seen (Fig. 7). We also examined the effect of blocking of each ligand and all ligands in 4 h cytotoxicity assay as described above. Consistent with surface ligand expression, blocking of ULBP-2 and ULBP-3 reduced cytotoxicity of $\gamma\delta$ T cells by 33 and 25% respectively while blocking of MIC-A/B and ULBP-1 had no effect of $\gamma\delta$ T cell lysis of U251MG cultures (Fig. 8). Blocking of all ligands had a small combined effect over that of blocking the individual ligands expressed by the cells (42%).

Discussion

We have previously shown that ex vivo expanded/activated $\gamma\delta$ T cells from a healthy donor can recognize and kill many GBM cell lines and primary tumor explants in a standard suspension cytotoxicity assay. In addition, we established that $\gamma\delta$ T cells are not cytotoxic to cultured normal astrocytes [18]. Treatment of human cell line xenograft tumors in immunodeficient mice provides the opportunity to test whether the therapeutic cell product can both migrate to and infiltrate the tumor following local/regional injection. In this report, we treated both minimal and established intracranial U251 human xenograft tumors with intracranial injection of expanded/activated $\gamma\delta$ T cells.

Anticipating future clinical trials, we adapted a cell manufacturing regimen originally developed by Lopez [27] for expansion and activation of $\gamma\delta$ T cells by substituting clinical-grade reagents for research-grade material and eliminating agents that cannot translate to clinical use (e.g. β-mercaptoethanol). In addition, we qualified the cell product for composition, purity, and potency using standard cGMP procedures, examples of which are shown in Figs. 1, 2, 3, 4. This regimen produces 100–600 fold expansion of activated $\gamma\delta$ T cells that contain both Vδ1 and Vδ2 subpopulations. They were able to produce a variety of immunomodulatory cytokines such as IFN-γ, TNF-α, and GM-CSF [30] and were able to kill U251MG cells in suspension and as an attached monolayer.

Since it is impractical to evaluate post-resection cell therapy in a mouse, we modeled therapy of residual microscopic GBM by injection of U251$^{\text{ffluc}}$ tumor cells and $\gamma\delta$ T cells at the same time. The U251 cell line was chosen as it has been shown in orthotopic xenograft models to be predictable and reliable in recapitulating the most salient pathobiologic features of GBM such as invasive behavior, pseudopalisading necrosis, robust angiogenesis, and caspase-3/HIF1α expression [31]. Our data demonstrate that expanded/activated $\gamma\delta$ T cells significantly inhibit tumor progression and improve survival in this model (Fig. 5). We then increased the stringency of this test by allowing U251$^{\text{ffluc}}$ tumors to grow for 7 days prior to treatment with $\gamma\delta$ T cells. The presence of tumor was confirmed by bioluminescence imaging between day +3 and +5 prior to treatment with $\gamma\delta$ T cells on day +7. Bioluminescence values varied widely between mice in all three groups and although mean bioluminescence trended lower for the mice that received $\gamma\delta$ T cells, the differences approached but did not achieve statistical significance for any time point. Median survival for the control group was 26 days following tumor injection, 32 days following a single injection of $\gamma\delta$ T cells, and 50 days in the group that received two injections of $\gamma\delta$ T cells. Interestingly, there were 4 long term disease-free survivors in the group that received one injection of $\gamma\delta$ T cells and 3 in the group that received two injections. One control mouse survived to the end of the study. Imaging performed prior to euthanasia revealed a slowly progressing tumor. These data suggest that at least in some cases expanded/activated $\gamma\delta$ T cells have the ability to migrate through brain parenchyma to locate and destroy U251 tumors. The potential contribution of tumor-derived immunosuppression of expanded/activated $\gamma\delta$ T cells to continued tumor progression remains mostly uncharacterized and will require further study. Technical factors may impact on survival as well, such as slight variations in graft quality and potency, distribution of the cell product within the brain, and migration of portions of the original tumor to a location less accessible to immune cells. Improved long-term survival of treated mice raises the possibility that a more aggressive dosing strategy may meet with greater success. Other than neurologic impairment that could be directly attributable to the presence of tumor, no obvious neurologic or systemic toxicities were observed as a result of the cellular therapy.

NKG2D ligands ULBP-2 and ULBP-3 were expressed on cultured cells and on solid tumor xenografts while MIC A/B and ULBP-1 expression was not seen (Fig. 7). Although generally consistent with flow cytometric data, the intensity of stress antigen expression, particularly ULBP-2, was variable in the solid tumors. This finding is not surprising considering the potential contribution of hypoxia, necrosis, and variable growth patterns of a three-dimensional tumor as compared with cultured cells. Our data indicate that the expression of stress associated ligands on this GBM cell line contributes to immunorecognition and lysis by expanded/activated $\gamma\delta$ T cells. Vδ1+ $\gamma\delta$ T cells recognize some NKG2D ligands via the T cell receptor [19, 32–35] and activated Vδ2+ T cells via NKG2D, both using mechanisms that are MHC-independent and require no prior antigen exposure or priming [33, 36, 37]. Blocking studies showed that the ligands ULBP-2 and ULBP-3 are factors in $\gamma\delta$ T cell recognition and cytolysis of U251 cells. Blocking of these ligands did not completely abrogate $\gamma\delta$ T cell cytotoxicity, likely illustrating the technical difficulties in maintaining a near 100% antigen block for 4 h at 37°C but also suggesting that these molecules alone are not the only ones involved in tumor immunogenicity to $\gamma\delta$ T cells. As many aspects of $\gamma\delta$ T cell function and potential ligand specificity remain unresolved [38], further studies will be necessary to define the interactions between GBM and $\gamma\delta$ T cells more completely.

Unfortunately, we were unable to document the presence of T cells in the brain parenchyma or tumor of mice, a problem that has also been reported by others [39] and possibly due to the tendency of activated $\gamma\delta$ T cells to undergo apoptosis following ex vivo culture [40] and persistent restimulation of the T cell receptor following tumor engagement [41]. Indeed, we have recently shown significant depletion of circulating $\gamma\delta$ T cells in newly-diagnosed patients with GBM [18], suggesting that design of future cellular therapies for GBM will likely have to incorporate both local placement of the cell product and repeated infusions.

In summary, we have shown, within the limits of the human/mouse tumor xenograft system, that local placement of ex vivo expanded/activated $\gamma\delta$ T cells appears to slow progression of established tumor resulting in improved survival. The application of T cell therapy based on innate recognition of stress-associated antigens for the treatment of malignant glioma provides a new and additional approach to augment anti-tumor immunotherapy in combination with currently available cytotoxic therapies.

Acknowledgments

This study was supported by NIH NCI Grant # P50 CA 097247, NINDS Grant # R21 NS 57341 and the National Brain Tumor Society Samuel Gerson Leadership Chair.

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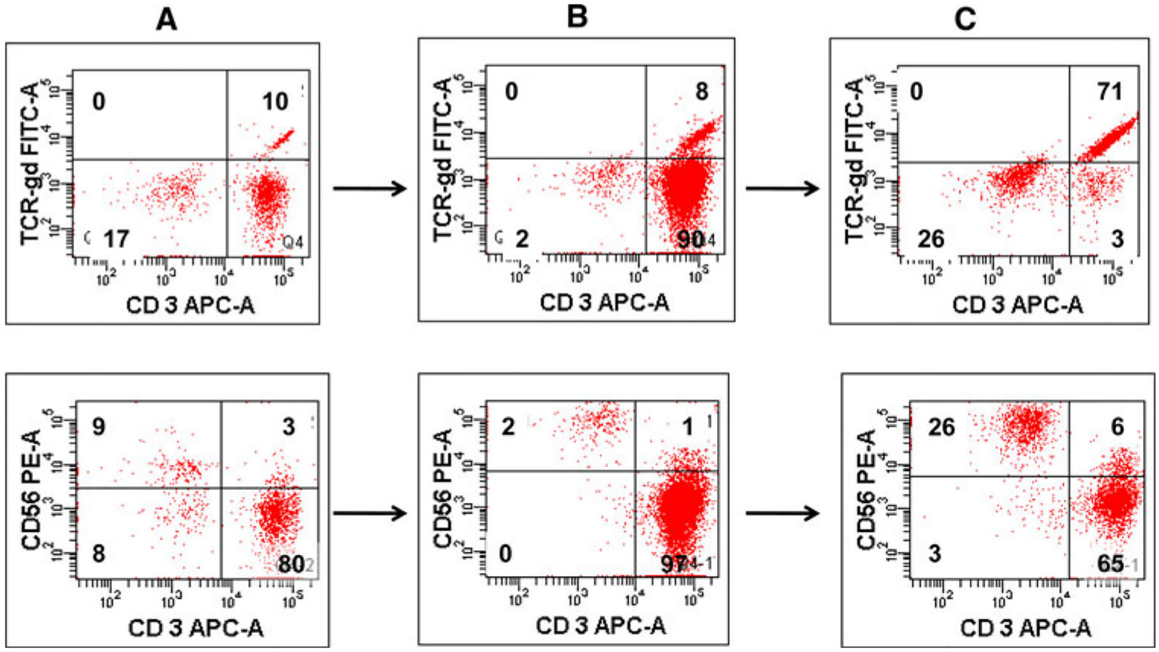

Fig. 1.

Representative example for manufacturing of $\gamma\delta$ T cell-rich cell therapy product using the CD2/OKT-3 procedure described in the text. The initial ficoll preparation from peripheral blood is shown in **a** and the 14-day expanded product is shown in **b**. This procedure expands both $\alpha\beta$ and $\gamma\delta$ T cells via OKT-3 and IL-2. Panel **c** represents the final product following depletion of most $\alpha\beta$ T cells using immunomagnetic microspheres coated with anti-CD4 and anti-CD8. From top left, $\gamma\delta$ T cells comprise 10% of the initial preparation, 8% of following 14 day expansion culture, and 71% of the final. The *bottom left* panels show that in this experiment, NK cells comprised 9.3% of lymphocytes in the starting preparation but only 2% of the 14 day expanded culture. NK cells are concentrated following depletion of CD4+ and CD8+ cells and comprise 26% of the final product

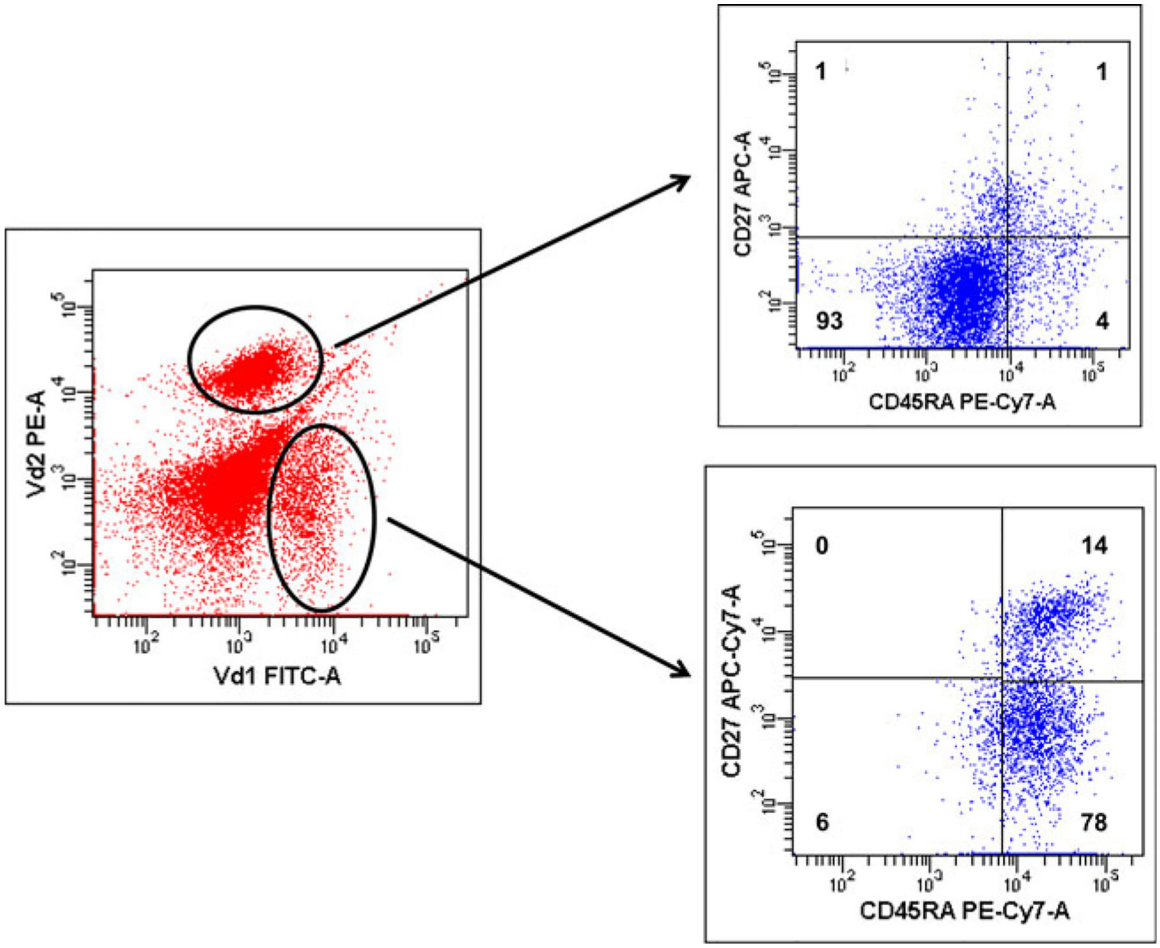

Fig. 2.
Distribution of Vδ1+ and Vδ2+ T cells following 14 day expansion and activation of $\gamma\delta$ T cells from the healthy donor shown in Fig. 1. The CD2/OKT3 procedure expands both major populations of $\gamma\delta$ T cells. Vδ1+ cells were predominately from the CD45+CD27− T-effector/memory RA phenotype (TEMRA) while Vδ2+ cells were effector/memory phenotype. A small population of naïve (CD45+CD27+) cells was seen in both Vδ1+ and Vδ2+ populations

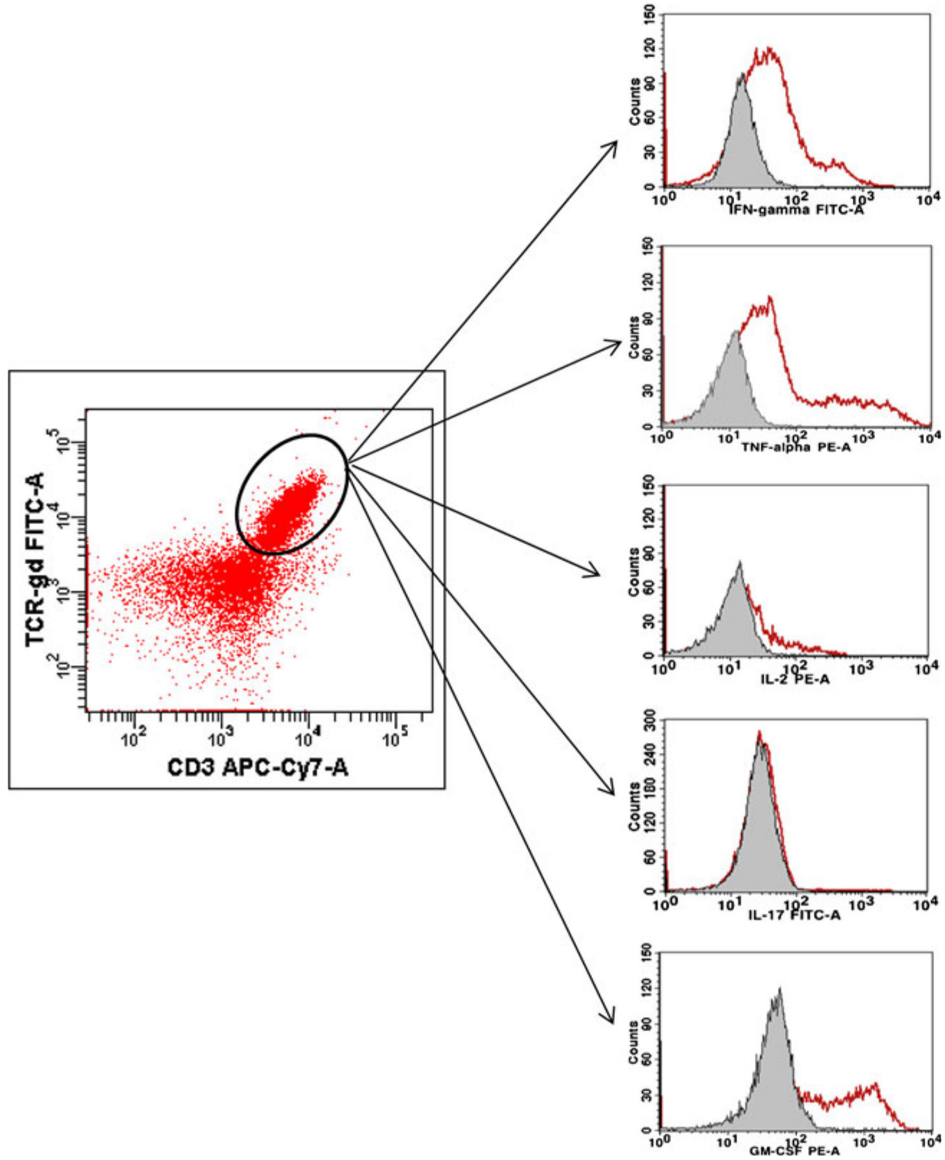

Fig. 3.

Intracellular cytokine staining of expanded/activated $\gamma\delta$ T cells from the healthy donor shown in Fig. 1. Note production of high levels of Th1 cytokines IFN-γ and TNF-α consistent with stimulation (see text)

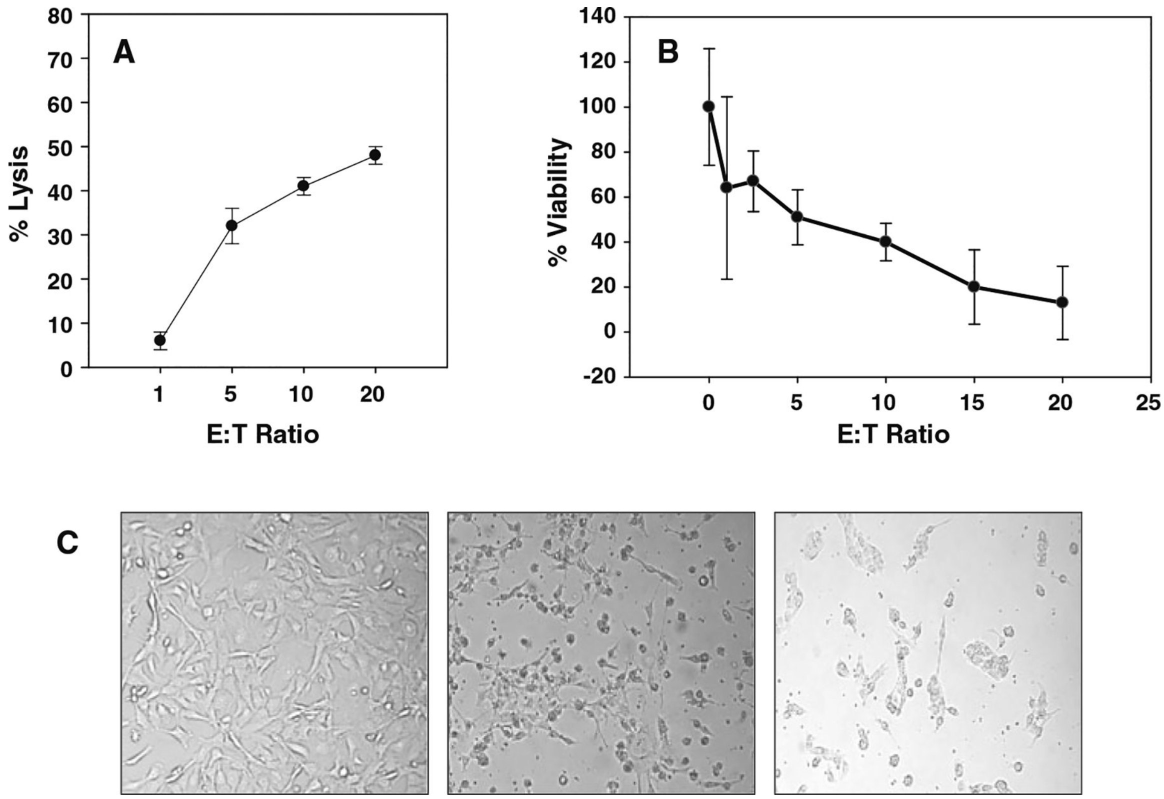

Fig. 4.

Anti-tumor activity of expanded/activated $\gamma\delta$ T cells demonstrated against human glioma cell line U251MG. **a** U251MG targets in suspension were labeled with ^{51}[Cr] and incubated with $\gamma\delta$ T cells for 4 h at the indicated effector to target ratios. Supernatants were removed to determine ^{51}[Cr] release in CPM. Data are presented at the mean percent target lysis (±SD) on triplicate determinations. **b** Viability of adherent U251MG cultures following 4 h incubation with expanded/activated $\gamma\delta$ T cells at the indicated effector to target ratios as determined by a commercial ATP release assay (see text). **c** From *left* to *right*, U251MG cells are shown in culture prior to addition of $\gamma\delta$ T cells and following culture with expanded/activated $\gamma\delta$ T cells for 4 and 24 h respectively

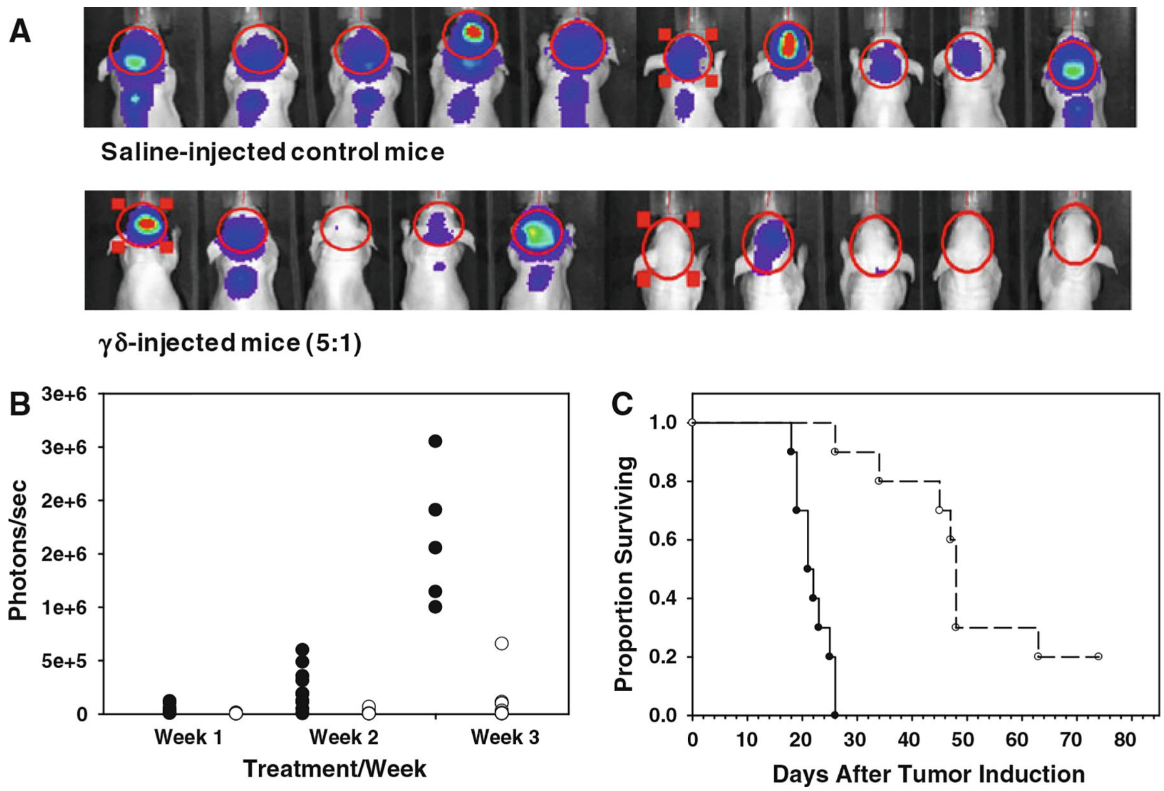

Fig. 5.

a Bioluminescence scan results at 14 days following tumor placement from 10 control mice injected with U251$^{\mathrm{ffluc}}$ GBM cells followed by saline (*top*) or $\gamma\delta$ T cells (*bottom*). Note that all 10 of the control mice have tumors present. However, 5 of the treated mice either failed to develop significant tumors or show significant tumor regression. **b** Photon emission data from the same experiment. Control mice receiving intracranial treatment with saline (*filled circles*) show more rapid progression of tumor than mice receiving $\gamma\delta$ T cell injections (*open circles*) over time. Each *circle* represents an individual mouse. **c** Survival plots of mice that received intracranial injections of saline (*closed circles*) and mice that received $\gamma\delta$ T cells (*open circles*) immediately following U251$^{\mathrm{ffluc}}$ tumor placement to simulate intraoperative post-resection therapy of residual GBM. Median survival of U251MG-injected mice treated $\gamma\delta$ T cells was 49 days with 2/10 75-day survivors as compared with control mice treated with saline ($P < .001$) and 0/10 75-day survivors

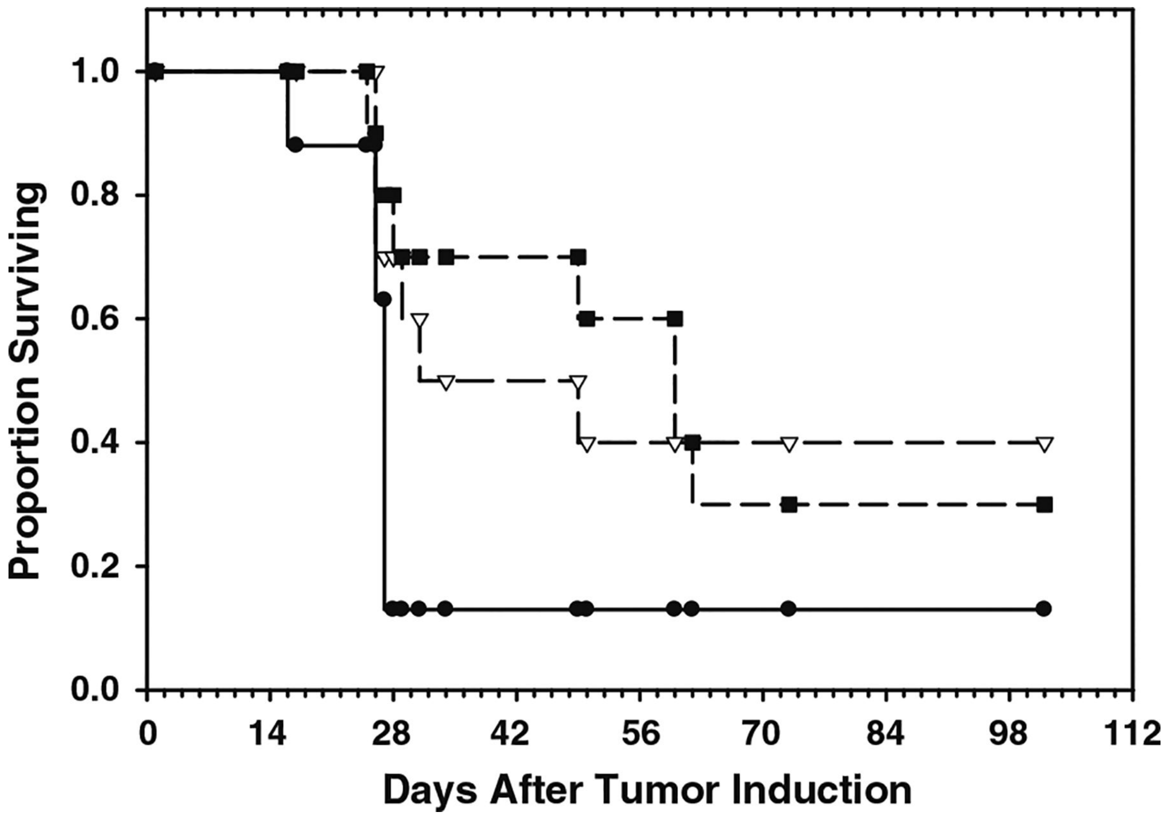

Fig. 6.

Survival plots of mice with established U251MG tumors. U251$^{\mathrm{ffluc}}$ tumors were placed 7 days prior to 2 weekly treatments consisting of saline/saline (*solid line*), $\gamma\delta$ T cells/saline (*long dashes*, *open diamonds*), or $\gamma\delta$ T cells/$\gamma\delta$ T cells (*short dashes*/*filled squares*). Median survival of U251MG-injected mice treated with one or two injections $\gamma\delta$ T cells weekly following tumor placement was 32 and 50 days respectively as compared 26 days for control mice

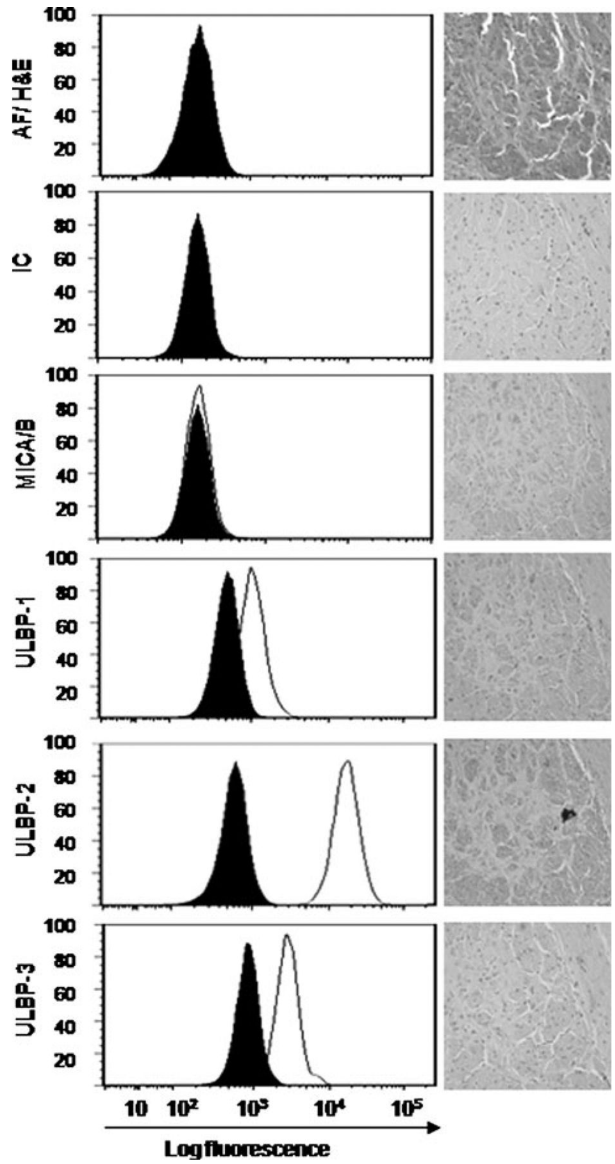

Fig. 7.

Expression of stress-associated molecules MIC-A/B, ULBP-1, ULBP-2, and ULBP-3 determined by flow cytometry on U251MG cells in suspension and by immunohistochemical staining on U251MG tumors removed from control mice. *Histograms* show autofluorescence (AF) on the *top panel* opposite H&E staining of the solid U251 tumor. Isotype controls are represented by filled histograms and specific antibody labeling by open histograms. Note that similar patterns of expression are present on both cells and tumors, although the intensity of labeling is variable within the tumor

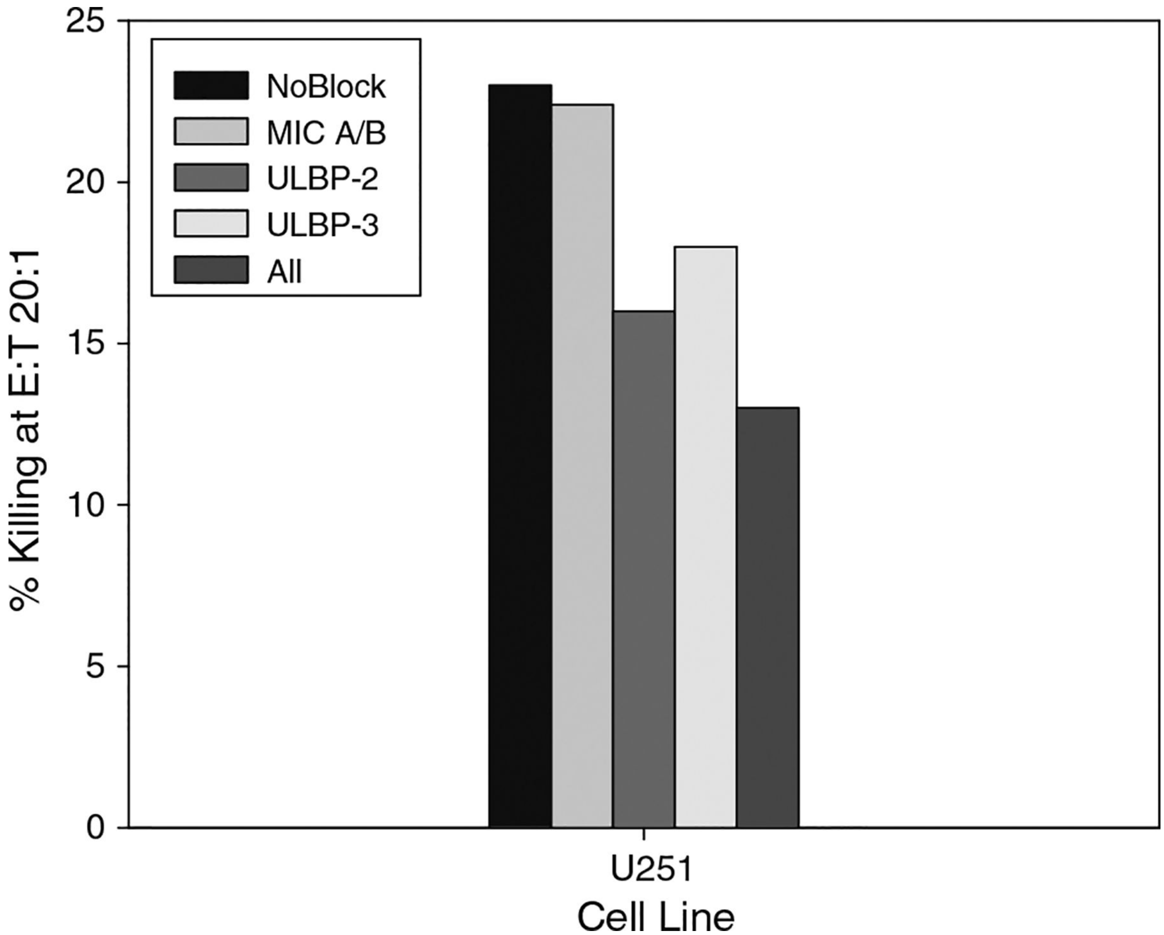

Fig. 8.

The stress antigens described in Fig. 7 were blocked on U251MG cells prior to addition of $\gamma\delta$ T cells at a 20:1 E:T ratio in a 4 h cytotoxicity assay. Note that neither blocking of any single stress antigen nor all antigens tested completely abrogated $\gamma\delta$ T cell cytotoxicity, suggesting the potential for additional ligands sensitive to $\gamma\delta$ T cell recognition